Establish a boat and fluid baseline

Task 3.2–3.3 / Prepare and draw

Name _____ Team _____ Date _____

01 Set up the investigation

Record the start rule and load. Speed = distance ÷ elapsed time. At the syringe station, mark initial piston positions; compare equal input strokes with air and water.

Mark start and finish lines. Record distance, time, load and test conditions for three comparable runs.

Boat run	Distance and units	Time (s)	Speed / load
1			
2			
3			

02 Plan it on paper

Predict what changes when linked syringes contain air instead of water. Mark the initial piston positions.

My prediction, chosen feature or design limit

Establish a boat and fluid baseline

Task 3.2–3.3 / Test, compare and explain

Name _____ Team _____ Date _____

03 Run the test and record what happened

Run the approved baseline boat at the teacher station. At the fluid station move the input piston a fixed distance and measure the output travel. Repeat each fluid trial. Look for bubbles, slipping tubing, or a changed load before interpreting differences.

Fluid and test	Input travel	Output travel	Units / leaks or bubbles
Air 1			
Air 2			
Water 1			
Water 2			

04 Use the reading

Ch. 18 | Shooting for the Moon | printed pp. 167-176
Complete the assigned checks in the interactive reader. Here, identify a page and one idea that helps explain your result or design decision.

05 Individual explanation

Give one baseline boat speed or a syringe travel comparison. Explain what you held constant and why.

Keep these sheets with the work. Save the individual explanation in the lesson’s Explain step.